HoneyBoo: Edge-Optimized Optical Flow and Computer Vision Pipeline for Bidirectional Entrance Monitoring and Temporal Activity Characterization in *Apis mellifera* Colonies

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Abstract*— Honeybees (*Apis mellifera*) serve as keystones for terrestrial ecosystems and global agricultural productivity. Conventional apiary assessment techniques necessitate intrusive hive disassembly, which disrupts internal thermoregulation, induces colony stress, and introduces mechanical mortality risks to the queen. Conversely, emerging Internet of Things (IoT) sensor arrays (internal microphones, scale balances) incur heavy capital expenditures and sensor propolization degradation. This paper presents HoneyBoo, an end-to-end, edge-optimized, non-invasive computer vision system for continuous, automated monitoring of honeybee entrance activity. Using raw monocular optical video acquired from standard consumer-grade smartphones or webcams without active illumination or auxiliary sensors, HoneyBoo combines an ultra-lightweight YOLOv8 spatial feature extractor with an adapted ByteTrack multi-object data association framework. By formalizing entrance crossing dynamics through 2D computational geometry and vector orientation cross-products, the system achieves bidirectional trajectory classification into incoming foragers and outgoing departures with an explicit triple-layer cooldown deduplication mechanism. Temporal traffic dynamics are modeled across rolling windows (1m, 5m, 15m), enabling statistical anomaly detection via rolling standard deviation Z-scores and differential rate-of-change metrics. Operating within strict computational boundaries (Intel Core i5, 8 GB RAM, 4 GB VRAM NVIDIA GTX 1650 or CPU fallback), HoneyBoo achieves 28.4 FPS throughput and a directional counting accuracy of 96.5% on verified benchmarks without cloud latency. Furthermore, the framework enforces an explicit scientific limitation boundary, decoupling entrance traffic observations from unverified internal colony disease diagnoses. Complete source code and models are open-sourced under the MIT license.

***Index Terms*— Computer Vision, Honeybee Monitoring, *Apis mellifera*, Multi-Object Tracking, ByteTrack, YOLOv8, Vector Cross-Product, Edge Computing, Precision Beekeeping.**

MANDATORY SCIENTIFIC LIMITATION NOTICE: *HoneyBoo detects and quantifies honeybee entrance movements using optical computer vision. Entrance activity indicators reflect behavioral traffic patterns and* ***must not be interpreted as confirmed colony disease diagnoses or comprehensive colony health assessments****. Entrance traffic fluctuations should always be corroborated by physical hive inspections.*

I. INTRODUCTION

The Western honeybee (*Apis mellifera*) is an indispensable pollinator supporting terrestrial biodiversity and facilitating the reproduction of over 75% of leading global food crops [1]. In recent decades, apiculturists have faced unprecedented colony losses driven by synergistic stressors, including habitat loss, parasitic mite vectors (*Varroa destructor*), pathogenic gut microsporidia (*Nosema ceranae*), agrochemical exposure, and unpredictable weather volatility [2], [3].

To evaluate colony vitality, beekeepers traditionally rely on periodic manual hive inspections. This manual methodology exhibits severe structural shortcomings: (1) Opening the hive envelope destroys the nest's precise 34.5°C ± 1.0°C internal thermoregulation, forcing substantial caloric reallocation toward nest reheating [4]; (2) Frame manipulation provokes defensive stinging, comb damage, and accidental queen injury or supersedure events [5]; and (3) Inspections are performed intermittently (typically every 7 to 14 days), failing to capture transient phenological crises, acute pesticide kills, or sudden foraging collapses [6].

Recognizing these limitations, the emerging discipline of Precision Beekeeping has explored automated, non-invasive bio-sensing [7]. While internal acoustic probes [8] and hive scales [9] offer macro-level signals, they suffer from propolization (bees encasing sensors in sticky antimicrobial resins), high hardware cost, and an inability to track directional forager flows.

Optical computer vision at the hive entrance provides an attractive, non-contact monitoring alternative. The entrance flight board acts as a natural bottleneck through which all foraging departures and incoming pollen/nectar imports must pass [10]. However, edge computer vision at hive entrances presents formidable algorithmic challenges: rapid flight velocities (>5 m/s) causing motion blur, severe spatial

occlusions during foraging rushes, lingering guard bees creating ambiguous oscillatory crossings, and the necessity to run locally on consumer-grade laptop hardware (Intel Core i5, 4 GB VRAM or CPU) without paid cloud subscriptions. To address these demands, this paper introduces **HoneyBoo**, an edge-optimized, end-to-end computer vision architecture designed for accurate bidirectional counting and temporal anomaly characterization.

II. RELATED WORK

Prior automated monitoring approaches span three technological paradigms: (1) Non-optical bio-acoustics and load cells, (2) Physical photoelectric entrance gates, and (3) Computer vision pipelines.

Ferrari et al. [11] investigated internal acoustic frequency spectra, identifying characteristic 300–500 Hz worker piping frequencies preceding swarming. Meikle et al. [12] used continuous scale balances to correlate daily mass fluctuations with net foraging gains. However, mass scales cannot distinguish between high worker mortality, water foraging, or food consumption, nor can they quantify individual bee counts. Early optical counters utilized mechanical IR beam-interruption tunnels [13], [14]. Unfortunately, physical tunnels create unnatural congestion, dislodge pollen pellets from returning foragers, and suffer from bee propolization.

In non-contact computer vision, early methods deployed classical background subtraction and optical flow [15], [16]. These methods fail under dynamic outdoor sunlight, moving shadows, and high bee clustering. Recent works transitioned to deep convolutional networks. Bjerge et al. [17] and Kulyukin et al. [18] applied YOLO and Faster R-CNN architectures to landing board imagery. Nevertheless, existing systems exhibit crucial deployment limitations: they either rely on computationally heavy models demanding expensive workstation GPUs (>16 GB VRAM) or lack robust multi-object tracking, relying on naive centroid proximity that fails during trajectory crossovers. HoneyBoo bridges this gap by coupling YOLOv8n with ByteTrack and a 2D vector cross-product crossing engine running on commodity laptops.

III. PROBLEM FORMULATION AND SYSTEM OBJECTIVES

Let a hive entrance video sequence be defined as an ordered set of digital frames $V = \{I_1, I_2, ..., I_T\}$, where $I_t \in R^{(H \times W \times 3)}$, with height H, width W, and discrete frame sampling rate f_s. At each time step t, the system must: (1) Detect all visible bees $D_t = \{d_t^1, ..., d_t^{(N_t)}\}$ with bounding boxes b_t^i and confidences c_t^i; (2) Maintain persistent trajectory identities $T_t = \{\tau^1, ..., \tau^M\}$ across visual occlusions; (3) Determine whether trajectory τ^k crossed virtual entrance line segment L = AB; (4) Classify the transit into directional event $E_t^k \in \{INCOMING, OUTGOING, \varnothing\}$; and (5) Aggregate crossings into temporal intervals to detect statistical anomalies within an inference latency $t_{infer} < 1/f_s$.

IV. METHODOLOGICAL FRAMEWORK & MATHEMATICAL FORMULATION

HoneyBoo operates via four pipelined computational stages: Spatial Rescaling, YOLOv8 Spatial Feature Extraction, Two-Stage ByteTrack Data Association, and Vector Cross-Product Crossing Classification.

A. Detection Architecture and Low-Resource Rescaling

To process high-resolution input videos within 4 GB VRAM without out-of-memory faults, incoming frames are dynamically rescaled:

$$s = \min(1.0, W_{target} / W), W_{target} = 960 \text{ px}, I'_t = \text{Resize}(I_t, (sW, sH)) \quad (1)$$

The object detection backbone utilizes **YOLOv8 Nano (YOLOv8n)**, an anchor-free convolutional architecture featuring Cross-Stage Partial with 2 Convolutions (C2f) modules and an independent decoupled head. The model optimizes a compound objective function:

$$L_{total} = \lambda_{box} L_{CIOU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL} \quad (2)$$

where L_{CIOU} enforces Complete Intersection over Union, L_{BCE} optimizes classification loss, and L_{DFL} represents Distribution Focal Loss for precise sub-pixel boundary regression. Detections are projected back to raw pixel space: $b_t = (1/s)\, b'_t$.

B. Multi-Object Data Association via Two-Stage ByteTrack

Because honeybees in rapid flight frequently experience partial wing occlusion, their detection confidence temporarily drops into the [0.15, 0.40) range. Standard trackers (e.g. SORT) discard detections below 0.50, generating broken trajectories. HoneyBoo adapts ByteTrack [19] using a two-stage association strategy:

$$D_{high} = \{d \in D_t \mid c_d \geq \theta_{high}\}, \theta_{high} = 0.40; D_{low} = \{d \in D_t \mid \theta_{low} \leq c_d < \theta_{high}\}, \theta_{low} = 0.15 \quad (3)$$

Active tracks are propagated via a constant velocity motion model with exponential velocity smoothing ($\alpha = 0.6$):

$$v_t = \alpha\, v_{(t-1)} + (1 - \alpha)\, (p_t - p_{(t-1)}) / \Delta t \quad (4)$$

1. High-confidence detections D_{high} are associated first with active tracks by minimizing an IoU distance cost matrix $C_{ij} = 1 - \text{IoU}(b_i, b_j)$.
2. Unmatched tracks are subsequently associated with low-confidence detections D_{low}, recovering occluded bees without false tracks.
3. New tracks require $N_{hits} \geq 3$ consecutive detections for confirmation; tracks lost for $N_{age} > 30$ frames are pruned.

C. 2D Vector Cross-Product Line Crossing and Directional Classification

The virtual entrance line is defined by two user-selected endpoints $a = (x_A, y_A)$ and $b = (x_B, y_B)$. Let tracked bee centroid coordinates across consecutive frames be $p_{prev} = (x_{(t-1)}, y_{(t-1)})$ and $p_{curr} = (x_t, y_t)$.

The signed perpendicular distance from an arbitrary point p to directed segment $a \rightarrow b$ is formulated via the 2D exterior vector cross-product:

$$S(p; a, b) = (b_x - a_x)(p_y - a_y) - (b_y - a_y)(p_x - a_x) \quad (5)$$

Geometrically, $S(p) > 0$ defines the left half-plane, $S(p) < 0$ defines the right half-plane, and $S(p) = 0$ denotes collinearity. A crossing occurs when the trajectory segment intersects line segment ab:

$$\text{Intersect} = [\text{sign}(S(p_{prev}; a, b)) \neq \text{sign}(S(p_{curr}; a, b))] \wedge [\text{sign}(S(a; p_{prev}, p_{curr})) \neq \text{sign}(S(b; p_{prev}, p_{curr}))] \quad (6)$$

To determine movement direction without hardcoding camera orientation, an interior probe point p_{probe} is projected offset into the designated hive interior side (e.g. +y for Bottom slot). Let $\sigma_{inside} = \text{sign}(S(p_{probe}; a, b))$. The crossing event is classified:

$$E_t^k = \text{INCOMING if } \text{sign}(S(p_{prev})) \neq \sigma_{inside} \wedge \text{sign}(S(p_{curr})) = \sigma_{inside}$$
$$E_t^k = \text{OUTGOING if } \text{sign}(S(p_{prev})) = \sigma_{inside} \wedge \text{sign}(S(p_{curr})) \neq \sigma_{inside} \quad (7)$$

D. Triple-Layer Cooldown and Anti-Duplication Filtering

Bees lingering or fanning near the entrance line are prevented from triggering duplicate counts via: (1) Requiring trajectory history length $L \geq 3$ points before crossing eligibility; (2) Trajectory locking: upon a valid crossing, subsequent checks for track ID k are suppressed for $N_{cooldown} = 25$ frames; and (3) Ambiguity rejection of lateral trajectories parallel to the entrance line.

V. STATISTICAL ACTIVITY DYNAMICS & TEMPORAL ANOMALY FORMULATION

Crossing events are aggregated into discrete temporal windows W_m (1m, 5m, 15m). Within each interval, total activity $C_{total}(m) = C_{in}(m) + C_{out}(m)$ and net movement $Net(m) = C_{in}(m) - C_{out}(m)$ are computed. To establish dynamic baselines under varying ambient conditions, rolling historical statistics across $K = 5$ intervals are maintained:

$$\mu_m = (1/K) \sum C_{total}(m - j), \sigma_m = \sqrt{[(1 / (K-1)) \sum (C_{total}(m - j) - \mu_m)^2]} \quad (8)$$

$$Z_m = (C_{total}(m) - \mu_m) / \sigma_m \quad (9)$$

• **HIGH_ACTIVITY:** $Z_m \geq +2.0$ (Activity exceeds the 97.7th percentile of rolling variance, signifying foraging departure waves or defense).

• **LOW_ACTIVITY:** $Z_m \leq -2.0 \wedge \mu_m \geq 3.0$ (Traffic collapses below baseline, indicating potential weather shutdown or disorientation).

• **SUDDEN_DROP:** $\Delta_m = (C_{total}(m-1) - C_{total}(m)) / \max(1, C_{total}(m-1)) \geq 0.60$ (Abrupt 60% drop in consecutive minutes).

VI. EXPERIMENTAL RESULTS AND EMPIRICAL EVALUATION

HoneyBoo was evaluated across two benchmark environments: physical apiary entrance recordings (10-minute field sequence at 1080p, 25 FPS with double-blind manual ground truth) and programmatic synthetic benchmarks with deterministic trajectories.

TABLE I: Detection Backbone Performance Comparison

Model	Params	mAP@0.5	mAP@0.5:0.95	Latency (GTX 1650)	VRAM Footprint
Faster R-CNN	41.5M	0.942	0.612	84.2 ms	3,420 MB
YOLOv8 Medium	25.9M	0.938	0.654	38.6 ms	2,180 MB
YOLOv8 Small	11.2M	0.924	0.631	18.2 ms	1,410 MB
YOLOv8 Nano (HoneyBoo)	**3.2M**	**0.916**	**0.618**	**8.4 ms**	**840 MB**

TABLE II: Multi-Object Tracking Evaluation on Honeybee Trajectories

Tracker Algorithm	Re-ID Network	MOTA (↑)	IDF1 (↑)	ID Switches (↓)	Tracking Speed
SORT [20]	None (Kalman)	74.2%	68.4%	89	112 FPS
DeepSORT [21]	MobileNet CNN	86.5%	84.1%	34	14 FPS
ByteTrack (HoneyBoo)	**Two-Stage IoU**	**88.9%**	**87.2%**	**18**	**82 FPS**

TABLE III: Directional Entrance Counting Verification

Test Evaluation Sequence	Ground Truth (In/Out)	HoneyBoo (In/Out)	Directional Acc.	False Positives	Net Drift Error
Synthetic Benchmark (8s)	3 / 3	3 / 3	100.0%	0	0
Field Trial A (Low Traffic)	42 / 38	41 / 37	97.5%	1	0
Field Trial B (Peak Foraging)	215 / 198	207 / 191	96.4%	5	-1
Field Trial C (Guard Hovering)	12 / 8	11 / 8	95.0%	1	0
Combined Aggregate	**272 / 247**	**262 / 239**	**96.5%**	**7**	**-1**

The system achieved an aggregate directional counting accuracy of **96.5%**. Counting errors predominantly occurred in dense flight clusters where multiple bees entered within an identical pixel boundary, causing brief detector bounding box mergers. Crucially, the net drift error was -1, confirming zero cumulative directional bias.

On the primary Intel Core i5 + GTX 1650 testbed, the complete end-to-end pipeline maintained a throughput of **28.4 FPS** with a VRAM footprint of 1,180 MB, easily accommodating 25 FPS real-time capture. In CPU fallback mode, throughput was 14.8 FPS; activating 2x frame skipping restored real-time operation at 29.6 effective FPS without tracking degradation.

VII. DISCUSSION & SCIENTIFIC LIMITATIONS

A. Biological Telemetry vs. Disease Diagnosis: HoneyBoo explicitly rejects claims that optical entrance video alone can diagnose internal colony diseases (e.g. American Foulbrood, Varroa infestation). Entrance traffic fluctuates naturally due to cloud cover, temperature drops, floral nectar flushes, and wind. Anomaly alerts represent behavioral observations that guide physical inspections, preventing agronomic misdiagnoses.

B. Physical Environmental Constraints: Prolonged direct rain requires a simple transparent weather shield. In exceptionally strong colonies (>60,000 bees) during swarming rushes, occlusion density can briefly terminate tracks. Finally, while bee crossings are quantified, pollen-carrying classification is left for future specialized multi-task heads.

VIII. CONCLUSION & FUTURE SCOPE

This research demonstrated that robust, bidirectional honeybee entrance monitoring can be achieved on commodity personal computers without specialized edge chips (Jetson/Pi) or paid APIs. By pairing YOLOv8n with ByteTrack, 2D vector cross-product geometry, and rolling Z-score anomaly indicators, HoneyBoo achieves 96.5% directional counting accuracy at 28.4 FPS. Future extensions will investigate multi-task pollen-load classification, behavioral action recognition (fanning, guarding), and micro-climate sensor integration.

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